Reading the Data: An Analysis of Willamette University Library's E-Resource Usage

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Abstract

WU Library subscribes to online journals from several publishers, and as Willamette's needs evolve, librarians must decide which electronic resources to keep and which to cut. Lacking a recent way to analyze usage patterns, we combined journal usage data with enrollment data to generate actionable, equitable suggestions for resource allocation. We found that usage ebbs when school is out of session, with Fall and Spring showing higher and more consistent usage. We also found that journal usage doesn't track closely with department enrollment, since journals are frequently used outside their subject area.

Introduction & Motivation

Problem & question: *Recently, the library has needed an avenue to understand the usage of the online resources they provide so they can effectively support student and staff needs in learning and research. This project explores usage patterns in regards to subject matter, semester, and department, and strives to make suggestions on which resources to prioritize. The final decisions on resource allocation will be decided by WU librarians.*

Impact & stakeholders: *This project could make impacts for the Willamette community as a whole, as the library's e-materials are an available resource for the everyone in the WU community. The largest stakeholder is the library staff, as they will have an avenue to evaluate the needs of students and the demand of various journals and their topics compared to the respective sizes of departments and majors.*

Background & Related Work

Previous Works: *The American Library Association (ALA) Data-Driven Librarianship toolkit, Institute of Museum and Library Services (IMLS) Public Library dashboard, South Central Library System in Wisconsin e-resource dashboard.*

This Project *intersects higher education contexts and enrollment data to develops a framework and initial data pipeline to store, analyze and make sense of Willamette's digital resource data and make data driven insights and decisions regarding digital resources.*

Methods & Evaluation

Approach: *We use exploratory analysis and descriptive statistics rather than predictive modeling, since the stakeholder question is descriptive: which resources are used, by whom, and how usage changes over time. Usage is measured with COUNTER Unique Item Requests, since Total Item Requests inflates differently across publisher platforms.*

Validation: *With no ground truth for "correct" usage, validation focused on data integrity. Two publishers supplied overlapping files, which we deduplicated before aggregating. A blank in a publisher report means "not reported," not zero, so the time series is restricted to journals reporting in every month.*

Results & Visualizations

Figure 1: Journal Usage by Month, 2023-2026

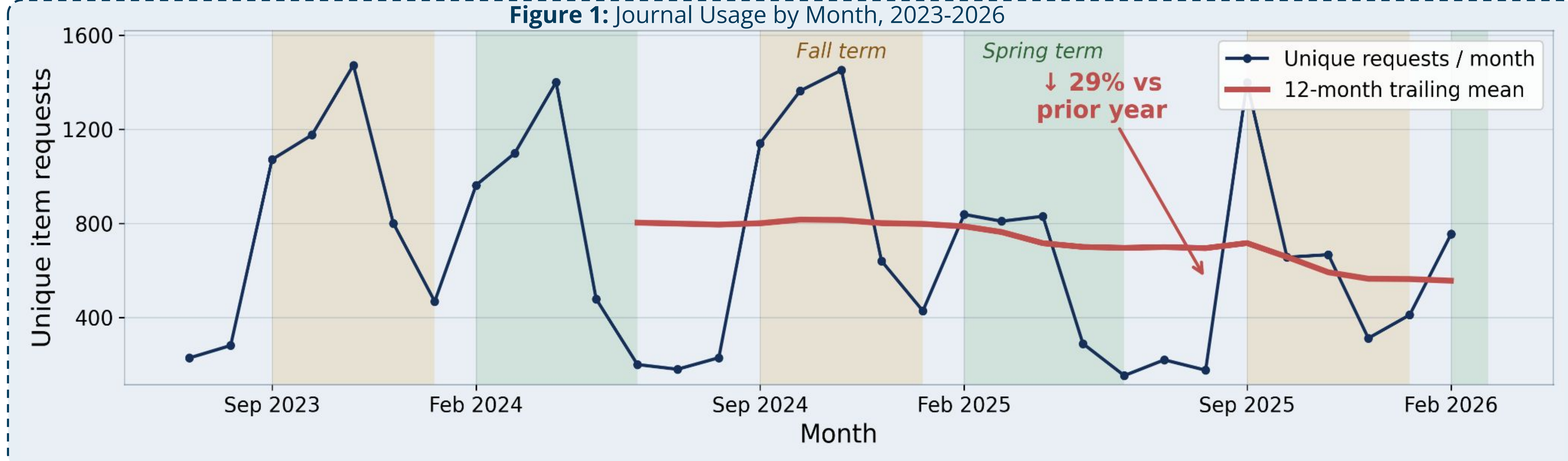
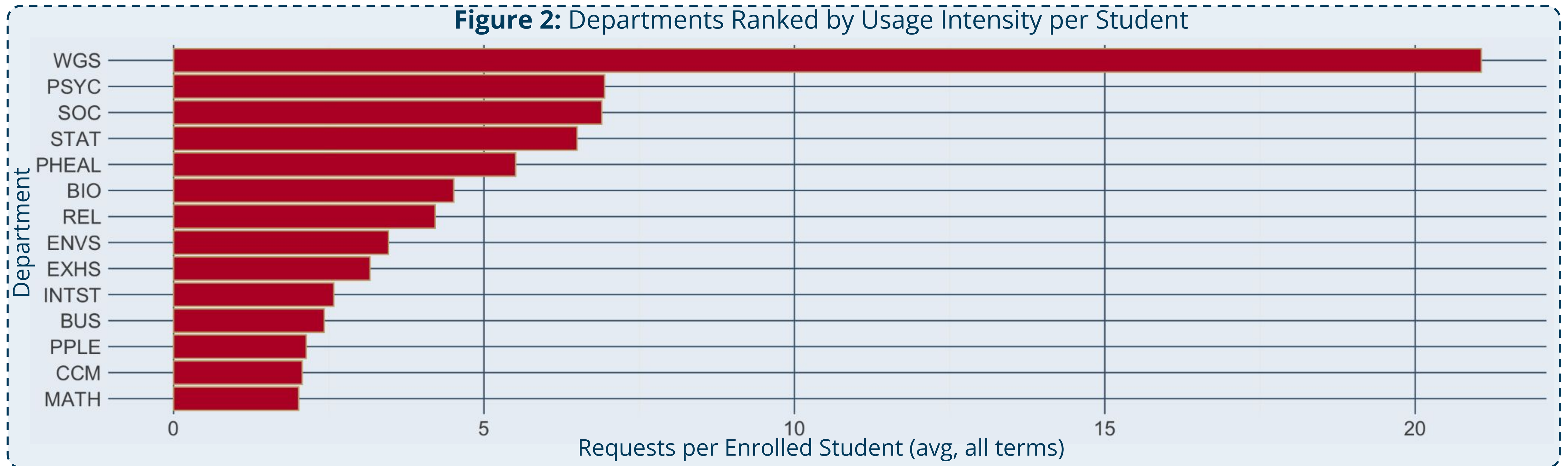


Figure 1 *shows monthly unique item requests from July 2023 to February 2026, for the 1,493 journals reporting in every month. March 2026 is excluded because only one publisher reported it. Usage follows the academic calendar, peaking mid-term and dropping to near zero over breaks, with Fall the highest of the three terms. Collection-wide usage declined 29% in the most recent year, from 787 to 556 unique requests per month. Because the journal panel is held constant and the decline appears within single-file publishers, it reflects real usage rather than a reporting gap. The isolated September 2025 spike is one newly activated journal, not a recovery. The seasonal timing is stable across all three years, so the decline is a change in level.*

Figure 2 *shows average journal requests per student by department per term. Women's & Gender Studies is a clear outlier, likely due to some mix of: popular topics in WGS journals, journal-intensive coursework, cross-listed classed using WGS-labeled journals, or use by other campus organizations. Psychology & Sociology follow, with 7-8 requests per student, which is consistent with their high journal volume. Both rank in the top 5 departments by journal count. STAT's 4th place ranking likely reflects statistical methods being used across many other disciplines. Overall, enrollment is not a strong indicator of journal usage, given this variation and the volume of out-of-major requests from students, faculty, and staff alike.*

Figure 2: Departments Ranked by Usage Intensity per Student

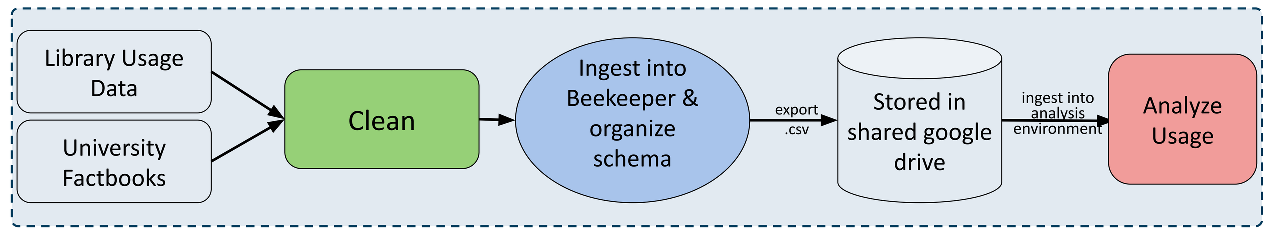


Data, Engineering & Ethics

Sources & scale: *The library usage data was given to us by the Willamette Library. It contains monthly usage for ~6000 journals from July 2023 - March 2026 from a few publishers. The enrollment data comes from the WU website. This includes enrollment for each major across all grades and semesters our data spans (Summer 2023-Fall 2025), this has ~400 rows of data.*

Ethics & responsible use: *All data is de-identified and contains only aggregate counts. As we do not want to cause harm to any departments, our findings are suggestions only. We strived to make equitable suggestions backed by our findings. Final decisions are made by the Willamette librarians.*

Pipeline:



Discussion, Limitations & Conclusions

Discussion: *Decreases in overall usage indicate that student e-resource usage is declining as a whole. Yet, usage varies based on the department, with Women's & Gender Studies (WGS) journals being utilized the most given enrollment size. Various reasons may explain these patterns including the Willamette's Gender Resource and Advocacy Center (GRAC) utilizing WGS resources, popularity of WGS courses at Willamette and the interdisciplinary nature of the topic.*

Limitations & future work: *This project analyzes the requests per student. The requests data includes journal access from everyone in the Willamette community, not just students potentially inflating usage for each discipline. Similarly, journals that are requested by out-of-major students and journals that don't fit neatly into one subject may also inflate usage patterns seen across majors. Future work would involve refining these categories, and doing contextual research to understand underlying reasons for declining usage in e-resources and variation in topic area journal requests.*

Conclusions: *The findings indicate clear patterns based on time and department. This offers the opportunity to continue exploring these variables and the underlying reasons to identify potential actions to address any disparities in usage or declines in usage. Greater understanding of underlying student research habits and topic variation will support in ensuring optimization of resources to support student needs at Willamette.*

Reproducibility & References

Repository: [https://github.com/sophiarabb/data510-libraries] · Key dependencies: [Python 3.x, pandas, scikit-learn, ...]

References: [1] American Library Association, Data-Driven Librarianship, www.ala.org/pla/data, 2026, [2] South Central Library System, Online Resources Usage Statistics by Library, https://www.scls.info/data-services/dashboards/online-resource-usage, 2026